

2021-2022 HS VCM #3 - Science

ANSWER KEY

B01. A	C01. D	P01. D
B02. C	C02. C	P02. B
B03. E	C03. D	P03. C
B04. D	C04. C	P04. A
B05. A	C05. C	P05. E
B06. D	C06. E	P06. E
B07. B	C07. E	P07. B
B08. C	C08. B	P08. A
B09. C	C09. E	P09. B
B10. A	C10. C	P10. E
B11. D	C11. C	P11. D
B12. B	C12. A	P12. E
B13. B	C13. B	P13. E
B14. C	C14. A	P14. C
B15. A	C15. C	P15. A
B16. D	C16. A	P16. B
B17. D	C17. D	P17. B
B18. C	C18. D	P18. E
B19. B	C19. D	P19. C
B20. D	C20. B	P20. D

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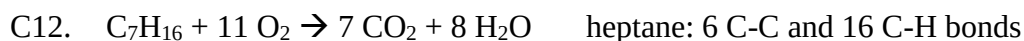
SELECTED SOLUTIONS

C01. $E = 2.178 \times 10^{-18} \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$ $wavelength = \frac{hc}{E} = 123 \text{ nm}$

C04. Molar mass = $(46.0/11.2)(0.0821)(273)/1 = 92 \text{ g/mol}$

C05. pH = 4.00 so $[H^+] = 10^{-4} \text{ M}$ $K_a = [H^+][A^-]/[HA] = (10^{-4})^2/1 - 10^{-4} = 10^{-8}$

C10. $\frac{70.0 \text{ g compd}}{100.0 \text{ g ore}} \left(\frac{190.65 \text{ g Cu}}{344.65 \text{ g compd}} \right) = 0.387 \frac{\text{g Cu}}{\text{g ore}} = 38.7 \% \text{ Cu}$



$$[(6)(348) + (16)(414) + (11)(498)] - [(14)(707) + (16)(464)] = -3132$$

C13. $\ln \left(\frac{P}{760} \right) = \frac{39300}{8.31} \left(\frac{1}{351.5} - \frac{1}{323.2} \right)$ $P = 234$

C17. $\Pi = (4.0)(0.0821)(298) = 98 \text{ atm}$ (NaCl = 2 moles particles/mole compound)

C19. $CaF_2(s) \rightleftharpoons Ca^{2+} + 2F^-$ $K_{sp} = [Ca^{2+}][F^-]^2 = (2.2 \times 10^{-4})(4.4 \times 10^{-4})^2 = 4.3 \times 10^{-11}$

P01-P03 The God Equation, Dr. Michio Kaku, Doubleday, New York, 2021

P01. page 13-14, a comet sailed over London in 1682. Edmond Halley traveled to Cambridge to meet with Isaac Newton who showed him that comets move in elliptical paths around the sun and that he could predict their trajectory. On Christmas Day in 1758, the comet sailed over Europe as predicted.

P02. page 47, when some were doubting Einstein's theory of relativity, Max Planck wrote to reassure Einstein. He stated, "A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because the opponents eventually die and a new generation grows up that is familiar with it."

P03. page 72-73, Joseph von Fraunhofer designed the most precise and accurate spectrograph of his time. He used this device to determine that the sun was mainly made of hydrogen. A new unknown substance was also found. It was named helium, which means "metal from the sun."

P04. 21st Century Astronomy, Laura Kay, Stacy Palen, Brad Smith, and George Blumenthal; W.W. Norton; 2013, page 270, "Above the tropopause and extending upward to an altitude of 50 km above sea level is the stratosphere."

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$$y = y_0 + v_0 t + .5at^2$$

$$0 = 150 + 0 + .5(-9.8)t^2$$

$$P06. t = 5.532833352$$

$$x = x_0 + v_0 t + .5at^2$$

$$x = 0 + 67t + 0 = 371 \text{ m}$$

$$P05. \frac{(500)(1000)}{(2.2)(12)^3(2.54)^3} = 8.0$$

$$\sum F_x = ma, F \cos \theta - F_f = ma$$

$$\sum F_y = 0, F_N + F \sin \theta - mg = 0$$

$$P07. F_f = \mu F_N, F_N = mg - F \sin \theta$$

$$(15) \cos(28.8^\circ) - (.345)[(3.25)(9.8) - (15) \sin(28.8^\circ)] = 3.25a$$

$$a = 1.43 \text{ m/s}^2$$

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_3$$

$$(5000)(12) + 0 = (11000) v_3$$

$$v_3 = 5.45, \text{ east}$$

$$P08. 0 + (6000)(10) = (11000) v_4$$

$$v_4 = 5.45, \text{ south}$$

$$v = \sqrt{v_3^2 + v_4^2} = 7.71 \text{ m/s, southeast}$$

$$\sum \tau = 0, \tau = rF \sin \theta$$

$$P09. (4.5)(12)(9.8) \sin(40^\circ) + (7)(40)(9.8) \sin(40^\circ) = (10)T \sin(50^\circ)$$

$$T = 275 \text{ N}$$

$$f' = f \left(\frac{v + v_o}{v} \right)$$

$$P10. f' = 1200 \left(\frac{343}{343 - 28.5} \right)$$

$$f' = 1310 \text{ Hz}$$

$$\Delta V = \beta V \Delta T$$

$$P11. (210 \times 10^{-6})(.842)(65.5) - (9 \times 10^{-6})(.842)(65.5)$$

$$0.0111 \text{ L} \rightarrow 11.1 \text{ mL}$$

$$Q = CV \left(1 - e^{\frac{-t}{RC}} \right)$$

$$P12. Q = (36 \times 10^{-3})(72) \left(1 - e^{\frac{-.048}{(288 \times 36 \times 10^{-3})}} \right)$$

$$Q = 0.0120 \text{ C} \rightarrow 12.0 \text{ mC}$$

$$E = \frac{kQ}{r^2}, E_1 = \frac{(8.99 \times 10^9)(558 \times 10^{-6})}{(.1106)^2}$$

$$E_1 = 410,094,209 \text{ N/C, right}$$

$$P13. E_2 = \frac{(8.99 \times 10^9)(766 \times 10^{-6})}{(.0862)^2}$$

$$E_2 = 926,774,188 \text{ N/C, left}$$

$$E_1 + E_2 = 5.17 \times 10^8 \text{ N/C, left}$$

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$$\text{P14. } F = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2 L}{d} = \frac{(2 \times 10^{-7})(6.48)^2 (24.4)}{.00882}$$

$$F = 0.0232 \text{ N}$$

$$B = \frac{k\mu_0 NI}{L}$$

$$\text{P15. } B = \frac{(588)(4\pi \times 10^{-7})(240)(6.44)}{.48}$$

$$B = 2.38 \text{ T}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\text{P16. } \frac{1}{-24} = \frac{1}{24} + \frac{1}{d_i}, \quad d_i = -12$$

$$M = -\frac{-12}{24} = \frac{1}{2}$$

virtual, erect, reduced

$$K = mc^2 \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right)$$

$$\text{P17. } K = (9.11 \times 10^{-31})(3 \times 10^8)^2 \left(\frac{1}{\sqrt{1 - (.752)^2}} - 1 \right) \left(\frac{1}{1.602 \times 10^{-19}} \right)$$

$$K = 265,000 \text{ eV} \rightarrow 0.265 \text{ MeV}$$

$$\text{P18. } \omega$$

$$\text{P19. } T = \sqrt{\frac{r^3}{r_{\text{Earth}}^3}}$$

$$138 = \sqrt{\frac{4000^3}{r^3}}$$

$$r = 150$$

$$\text{P20. } C = \frac{k\varepsilon_0 A}{d}$$

$$160 \times 10^{-9} = \frac{9(8.854 \times 10^{-12})(2.22)}{d}$$

$$d = 0.00111 \text{ m} \rightarrow 1.11 \text{ mm}$$

The Virtual Challenge Meets Science Contest Answer Sheet

Conference _____

Grade Level _____

Contestant # _____

Biology

B01 _____

B02 _____

B03 _____

B04 _____

B05 _____

B06 _____

B07 _____

B08 _____

B09 _____

B10 _____

B11 _____

B12 _____

B13 _____

B14 _____

B15 _____

B16 _____

B17 _____

B18 _____

B19 _____

B20 _____

B Score

Chemistry

C01 _____

C02 _____

C03 _____

C04 _____

C05 _____

C06 _____

C07 _____

C08 _____

C09 _____

C10 _____

C11 _____

C12 _____

C13 _____

C14 _____

C15 _____

C16 _____

C17 _____

C18 _____

C19 _____

C20 _____

C Score

Physics

P01 _____

P02 _____

P03 _____

P04 _____

P05 _____

P06 _____

P07 _____

P08 _____

P09 _____

P10 _____

P11 _____

P12 _____

P13 _____

P14 _____

P15 _____

P16 _____

P17 _____

P18 _____

P19 _____

P20 _____

P Score

Grader Initials _____

OVERALL SCORE